# Heteroscedasticity, Autocorrelation, Outliers & Leverage Points, Effect Size, and Confidence Intervals

**Hetroscedasticity**

Hetroscedasticity occurs when the variance of regression errors is not constant across different levels of the predictor or predicted outcome.

In simple words, a model;s prediction errors may be small for some observations but much larger for others.

**Technically:**

It occurs when the conditional variance of the regression error varies with the predictors:

$$Var(\varepsilon_i | X_i) = \sigma_i^2$$

If the error variance is contstant, the assumption is called **homoscedasticity.**

Example:

Retailer predicts sales using ad expenditure.

1. Store spending ₹10,000 on advertising might have prediction errors of around ₹1,000.
2. Store spending ₹10 lakh might have prediction errors of ₹1 lakh.

Homoscedasticity

Risidual spread stays approximately constant

Heteroscedasticity

Risidual spread becomes wider or narrower.

| Store | P_Sales | Actual_Sales | Residual(Y-Ŷ) | e² |
|---|---|---|---|---|
| A | 10 | 11 | 1 | 1 |
| B | 20 | 18 | -2 | 4 |
| C | 30 | 33 | 3 | 9 |
| D | 40 | 36 | -4 | 16 |
| E | 50 | 56 | 6 | 36 |
| | | | MSE | 13.2 |

For store E:

$$MSE = \frac{\Sigma e_i^2}{n}$$

e = Y-Ŷ = 56-50 = 6

e² = 36

The residual magnitude grows from 1 to 6 in this dataset. This is a resaon to investigate whether erros become more variable at higher predicted sales, although 5 obseravtions alone cannot establish heteroscedasticity.

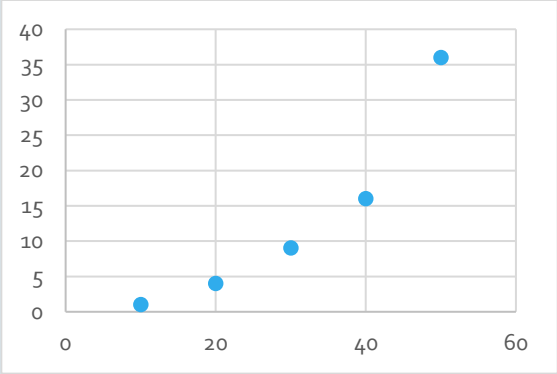
**Detection and solution:**

Methods to investigate:

1. Residual-vs fitted plot
2. Breusch -Pagan test
3. White test
4. Review of residual spread across business segments.

Possible responses:
1. Use heteroscedasticity-robust standard erros when appropriate.
2. Consider a log transformation for positive, right skewed outcomes
3. Add relevant ommited variables.
4. Reconsider the functional form of the model
5. Investiagte whether different groups need different models.

| P_Sales | $e^2$ |
|---|---|
| 10 | 1 |
| 20 | 4 |
| 30 | 9 |
| 40 | 16 |
| 50 | 36 |




**Business Application:**
As Sales foercasting model may perform adequately for small stores but have much larger errors for high volume stores. Analysts should check this before using standard refression confidence intervals without adjustment.

**Autocorrelation**
Autocorrelation occurs when observations, or regression errors, are related to their own past values.
It is especially important in time-series data, where observations are ordered by time.

Example:
If a retail store has unexpectedly high sales this month, sales next month may also be unusually high because of a continuing promotion or a season demand.

Autocorrelation in Time Series: Values are related to earlier values.
Autocorrelation of regression errors: Model errors are related across time potentially violating the independence assumption.

**Positive and negative autocorrelation**
+ve: High values tend to follow high values and low values tend to follow low values
-ve : High values tend to follow low values and vice versa

| Month | R( e ) |
|---|---|
| January | 2 |
| February | 3 |
| March | 4 |
| April | 3 |

Lets calculate a smile lag -1 residual correlation

Lag -1 pairs

| Current | Previous |
|---|---|
| 3 | 2 |
| 4 | 3 |
| 3 | 4 |

The mean of the paired colums are:

$$\bar{x} = \frac{2+3+4}{3} = 3 \qquad \bar{y} = \frac{3+4+3}{3} = \frac{10}{3}$$

Using pearson correlation forumal for these pairs:

$$r = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma (x_i - \bar{x})^2 \Sigma (y_i - \bar{y})^2}}$$

(-1)(-1/3) + (0)(2/3) + (1)(-1/3) =0

r =0 $\quad Corr(Y_t, Y_{t-1}) = \frac{Cov(Y_t - Y_{t-1})}{\sigma_{Y_t} \sigma_{Y_{t-1}}}$

**Detection and practical solutions:**

Methods:

Residual time series plot

Autocorrelation function(ACF)

Durbin-Watson test for first order residual autcorrelation in suitable regression settings

For Durbin Watson Statistic:

$$DW = \frac{\Sigma_{t=2}^{n} (e_t - e_{t-1})^2}{\Sigma_{t=1}^{n} e_t^2}$$

DW = 2: Little first order autocorrelation indicated

DW < 2 : Possible positive autocorrelation

DW >2: Possible negative autocorrelation

**Applications**

Monthly sales forecasting

Daily stock or demand data

Call center volume prediction

Production and inventory planning

**Outliers & Leverage points**

**Outlier:**

An observation that is unusually far from the general pattern of the data or from the fitted model.

large residual(+ve or -ve)

**Leverage Point:**

An observation with an unusual combination of predictor values, far from the center of the predictor data.

Unsual predictor value

**Influential Observation:**

A point whose inclusion or removal materially changes the fitted model.

Stong impact(effect) on fitted model

**Example:**
Suppose a company models sales against ad spend

| Store | Ad Spend(X) | Sales (Y) | |
|---|---|---|---|
| A | 1 | 3 | 3 |
| B | 2 | 5 | 5 |
| C | 3 | 7 | 7 |
| D | 4 | 9 | 9 |
| E | 5 | 11 | 11 |
| F | 10 | 5 | 21 |

$Y = 1 + 2X$

At X = 10, the predicted values based on the pattern is 21
But store F has actual sales of 5
Its residual relative to the first five store pattern is

$e = Y - \hat{Y} = 5 - 21 = -16$

So store F has an unusually large negative residual relative to the pattern formed by the five stores, and its advertising spend is also far beyond their range.
It is a candidate for both an outlier and a high -leverage point. Whether it is influential must be checked by comparing model fits or using influence diagnostics.

**What should analyst investigate?**
1. Was the advertising value entered correctly?
2. Did the store experience a stockout?
3. Was there a competitor promotion?
4. Is this store operating under a different business model?
5. Does the obseravtion represent a genuine but unsual event?

**Detecting Outliers using IQR rule**
10 12 13 14 15 18 20 22

Lower half:
10 12 13 14
Q1 = 12+13/2 = 12.5

Upper half
Q2 = 18+20/2 = 19

IQR = Q3-Q1 = 19-12.5 = 6.5

Lower fence 12.5 -1.5(6.5) = 2.75

Upper fence
19 + 1.5(6.5) =28.75
Therefore any value below 2.75 or above 28.75 is flagged as potential outlier under this rule.

**Leverage and Influential Diagnostics**

1. Standardized/ studentized residuals: Identify the observations with unusually large errors
2. Leverage (hat values): Identify the unusual predictor combinations.
3. Cook's Distance: Asses how much the fitted model changes when an observation is removed.
4. DFFITS: Measures an observation's influence on its own fitted value.

**Effect Size**

Effect size tells how large or practically important a difference or relationship is.

A p-value helps asses compatibility with a statistical null hypothesios. Effect size helps describe the magnitude of a difference or relationship.

**Cohen's *d*: Comparing two groups**

It measures the difference between two group means in standard devaition units.

$$d = \frac{\bar{X}_1 - \bar{X}_2}{sp}$$

Where,

$$sp = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

**Example:**

Sales Training

| Statisitic | Group A | Group B |
|---|---|---|
| Mean | 70 | 78 |
| SD | 10 | 10 |
| Sample Size | 25 | 25 |

$$sp = \sqrt{\frac{24(10)^2 + 24(10)^2}{48}}$$

$$sp = \sqrt{\frac{4800}{48}} = \sqrt{100} = 10$$

$$78 - 70 = 8$$

$$d = \frac{8}{10} \qquad \boxed{d = 0.80}$$

Group B's mean is 0.8 pooled standard deviations higher than group A's mean

**Question?**
Is the difference of 8 points enough to justify the training cost? Effect size helps quantify the difference, but the costs and uncertainty must be considered.

**Correlation effect Size and R square**

$$r = 0.60$$
$$R^2 = r^2 = 0.60^2 = 0.36$$
$$\boxed{R^2 = 0.36}$$

**Practical vs Statistical significance**

**Confidence intervals for regression coefficients**

A confidence interval gives a range of plausible values for a populaiton parameter based on sample data and specified method.

$$\alpha = 0.05 \quad CI = 95\%$$

For regression Coefficient:

$$\hat{\beta}_j \pm t_{\alpha/2, df} \, SE(\hat{\beta}_j)$$

ERC

**Example:**
An analyst estimates the relationship between Ad spend and sales
Estiamted slope b =2.5
SE of slope SE(b) =0.8
Sample Size n = 20
Number of predictors k = 1
CI = 95%
DF = n-k-1 =20-1-1= 18
The 95% two sided critical t value for 18 df is approximately t* = 2.101

ME = t* x SE(b)
ME = 2.101(0.8) = 1.6808

Lower limit
2.5-1.6808 = 0.8192
Upper limit
2.5+1.6808= 4.1808

95% CI = (0.819,4.181)

**Interpretaion:**
If advertising Spend increases by one unit, the model estimates an average increase of 2.5 sales units
The 95% CI for the sloope is approximately 0.819 to 4.181 sales units per advertising unit, under the model assumption.

This is evidence of association, not a proof that advertising causes the increase.

**Confidence interval for the intercept**

Estimated intercept a = 10

SE (a) =2

T* = 2.101.

ME = 2.101(2) =4.202

CI

$10 \pm 4.202$

$95\%\ CI = (5.798, 14.202)$

The estimated intercept is 10 with a 95% CI from approcimately 5.798 to 14.202

What is Cook's Distance?

What does cohen's d = 0.8 mean?

Does a statistically significant result always have a large business impact?